

Phase 9 Acceptance Record: Adviser Redaction Review Gate

Execution Date: 2026-09-02

Assessor: Automated Rigorous Test Suite (scripts/test-phase9.mjs)

Status: PASSED (46/46 Tests, 100% Compliance)

1. Scope & Invariants Evaluated

2. Test Execution Log Output

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CASE ACE v2.0 - PHASE 9: ADVISER REDACTION REVIEW GATE TESTS
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--- TEST 1: Session Initialization & Gate State Invariants ---
| PASS: VolatileSessionStore is clean before session init
| PASS: isGatePassed is initially false
| PASS: gateOpenedTimestampMs is initially null
| PASS: gateCompletedTimestampMs is initially null
| PASS: acknowledgedLowConfidenceIds is initially empty
| PASS: manualRedactions is initially empty

--- TEST 2: Low-Confidence Acoustic Escalation (<0.70) ---
| PASS: gateOpenedTimestampMs is recorded upon opening gate
| PASS: Session stage transitioned to redaction_review
| PASS: Extracted 3 low-confidence acoustic items
| PASS: checkGateReadiness returns canProceed: false when low-confidence items exist
| PASS: Pending count is 3 (blocking unlock)
| PASS: Blocking reason message is clearly articulated

--- TEST 3: Strict Blocking & Failsafe Invariant Verification ---
| PASS: unlockGate() throws descriptive error on unreviewed items
| PASS: volatileSessionStore.unlockGate() strictly blocks execution
| PASS: executeAffirmativeProceed(false) returns success: false
| PASS: Returns affirmative declaration error
| PASS: executeAffirmativeProceed(true) fails when items are pending
| PASS: Returns low confidence blocking error

--- TEST 4: Individual Acknowledgement of Low-Confidence Regions ---
| PASS: Gate still blocked after acknowledging only 1 of 3 items
| PASS: Acknowledged count is 1
| PASS: Pending count decremented to 2
| PASS: Gate readiness is now TRUE after all items individually acknowledged
| PASS: Pending low-confidence count reached strictly 0

--- TEST 5: Adviser Manual Redactions & False-Positive Removal ---
| PASS: Manual redactions array has 1 item
| PASS: Token map contains surrogate mapping for manual item
| PASS: Identifier decision marked as rejected
| PASS: Rejected surrogate token removed from active tokenMap

--- TEST 6: Outbound Transmission Disclosure & Affirmative Unlock ---
| PASS: Target region is europe-west2 London
| PASS: Credential validity is 300s ephemeral
| PASS: Outbound tokenised payload preview generated
| PASS: Surrogate substitution correctly reflected in outbound preview
[SecurityTelemetry] redaction_gate_completed {
  dwellTimeMs: 55,
  lowConfidenceReviewedCount: 3,
  manualAddedCount: 1,
  manualRemovedCount: 1,
  totalTokenCount: 3
}
| PASS: Affirmative proceed succeeds
| PASS: Active gate dwell time measured (55ms)
| PASS: isGatePassed is true in volatile state
| PASS: Session stage moved to tokenisation
| PASS: Redaction review audit record attached to session
| PASS: Audit record contains non-zero dwell time
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--- TEST 7: Security Telemetry Event Verification ---
' PASS: Telemetry buffer received 1 event(s)
' PASS: Security telemetry recorded "redaction_gate_completed" event
' PASS: Event details contain dwellTimeMs
' PASS: Event details contain lowConfidenceReviewedCount
' PASS: Zero client name in telemetry log
' PASS: Zero NINO in telemetry log
' PASS: Zero postcode in telemetry log
' PASS: Zero landlord name in telemetry log
' PASS: Zero transcript payload key in telemetry log
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PHASE 9 VERIFICATION SUMMARY: 46/46 TESTS PASSED
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3. Formal Sign-Off

Phase 9 (Adviser Redaction Review Gate) satisfies all specifications, security constraints, and automated tests. The system is ready for Phase 10 (Cloud Speech-to-Text v2 & Vertex AI Gemini Drafting Pipeline).